

City University of Science and Information Technology, Peshawar

Semester	Spring-2026	Examination	Final Term
Course Title	Web-Eng	Instructor	Kazim Ullah
Shift - Program(s)	Morning	Total Marks	100
Time Allowed: 60 Minutes			
Note: Attempt all questions.			

Final Term Lab Project Scenario

Library Book Management System for Web Engineering Course

You have been hired as a **web developer** by **City University Peshawar – Computer Science Department** to create a **Library Book Management System** for internal use in the Web Engineering Lab.

The system will help the librarian **add, view, update, and delete book records** without using external tools like Excel or manual registers.

Your task is to develop a **fully functional web-based CRUD application** using **HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL** within **60 minutes**.

The system should run locally using **XAMPP** or any local server environment.

Functional Requirements

Task 1. Database Setup

Marks: 20

- The application must use a MySQL database named:
 - yourname_cusitid
 - Create a table named books with the following fields:
 - title
 - author
 - price
-

Task 2. Add New Book

Marks: 20

- Provide an HTML form where the librarian can enter book details.
 - Apply validation:
 - Title should contain only alphabetic characters and spaces
 - Author name should contain only alphabetic characters
 - Price should be a number greater than 0
 - On successful submission, store data in the database.
-

Task 3. View All Books

Marks: 20

- Display all book records in a **Bootstrap-styled table**
 - Each row should include:
 - Book Title
 - Author
 - Price
 - **Edit** and **Delete** buttons
-

Task 4. Edit Book Record

Marks: 20

- Clicking **Edit** should load the selected book's data into a form
 - Allow the librarian to update book information
 - Save updated data to the database
-

Task 5. Delete Book Record

Marks: 20

- Allow the librarian to delete a book using the **Delete** button
 - Redirect back to the book listing page after deletion
-

Activity 1: Setting Up the Database

Open **phpMyAdmin** and run the following SQL command:

```
CREATE TABLE books (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    title VARCHAR(100),  
    author VARCHAR(100),  
    price DECIMAL(10,2)  
);
```

db_connect.php

```
<?php  
$servername = "localhost";  
$username = "root";  
$password = "";  
$dbname = "your_database_name"; // Replace with your DB name  
  
$conn = new mysqli($servername, $username, $password, $dbname);  
  
if ($conn->connect_error) {  
    die("Connection failed: " . $conn->connect_error);  
}  
?>
```

Activity 2: Creating a Form to Add Records

File: add_book.php

```
<form action="insert.php" method="POST">  
    <input type="text" name="title" placeholder="Enter book title" required>  
    <input type="text" name="author" placeholder="Enter author name" required>  
    <input type="number" name="price" placeholder="Enter price" step="0.01" required>  
    <button type="submit">Add Book</button>  
</form>
```

File: insert.php

```
<?php  
include 'db_connect.php';  
  
$title = $_POST['title'];  
$author = $_POST['author'];  
$price = $_POST['price'];  
  
$sql = "INSERT INTO books (title, author, price)
```

```
VALUES ('$title', '$author', '$price');"

$conn->query($sql);

header("Location: view_books.php");
?>
```

Activity 3: Displaying Records

File: view_books.php

```
<?php
include 'db_connect.php';

$sql = "SELECT * FROM books";
$result = $conn->query($sql);

while ($row = $result->fetch_assoc()) {
    echo $row['title'] . " - " . $row['author'] . " - " . $row['price'];
    echo " <a href='edit.php?id=" . $row['id'] . "'>Edit</a> | ";
    echo " <a href='delete.php?id=" . $row['id'] . "'>Delete</a><br>";
}
?>
```

Activity 4: Updating Records

File: edit.php

```
<?php
include 'db_connect.php';

$id = $_GET['id'];
$sql = "SELECT * FROM books WHERE id = $id";
$result = $conn->query($sql);
$row = $result->fetch_assoc();
?>

<form action="update.php" method="POST">
    <input type="hidden" name="id" value="<?php echo $row['id']; ?>">
    <input type="text" name="title" value="<?php echo $row['title']; ?>" required>
    <input type="text" name="author" value="<?php echo $row['author']; ?>" required>
    <input type="number" name="price" value="<?php echo $row['price']; ?>" step="0.01" required>
    <button type="submit">Update Book</button>
</form>
```

File: update.php

```
<?php
include 'db_connect.php';

$id = $_POST['id'];
$title = $_POST['title'];
$author = $_POST['author'];
$price = $_POST['price'];

$sql = "UPDATE books
      SET title='$title', author='$author', price=$price
      WHERE id=$id";

$conn->query($sql);

header("Location: view_books.php");
?>
```

Activity 5: Deleting Records

File: delete.php

```
<?php
include 'db_connect.php';

$id = $_GET['id'];
$sql = "DELETE FROM books WHERE id = $id";

$conn->query($sql);

header("Location: view_books.php");
?>
```
